

THEODORUS WENSAN FEBRUANTO

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EDUCATION

Arizona State University (Tempe, AZ)

Ph.D. in Computer Science, Aug. 2024 - Expected Aug. 2029

Advisor: Adil Ahmad

Research: Software Engineering and Systems Security

Thesis Question: How can LLMs be combined with traditional system analysis to transform human-readable system knowledge into machine-checkable semantics for automated system validation and security analysis?

Institut Teknologi Sepuluh Nopember (ITS) (Surabaya, Indonesia)

B.Eng. in Computer Engineering, 2019 - 2024

GPA: 3.54/4.00

RESEARCH

Specification-Guided Detection of ISA Conformance Bugs in RISC-V Emulators - *Under review at ICSE 2027*

Role: First Author | Arizona State University | Aug. 2024 - Present

- **Designed OculusISA**, an LLM-assisted, specification-guided framework that extracts architectural requirements from natural-language RISC-V ISA manuals, producing **11,876 specification rules** covering 241 registers and 1,173 instructions.
- **Implemented cross-privilege testing infrastructure** using xv6 and Linux across M, S, U, VS, and VU modes, with automated execution and oracle checking for **QEMU, Spike, RVVM, and Renode**.
- **Discovered 44 previously unknown ISA conformance bugs** across QEMU, RVVM, and Renode; **39 (88.6%)** have since been confirmed or fixed upstream.

RISolate: Automated Policy Derivation and Testing for RISC-V Context-Switch Isolation

Role: First Author | Arizona State University | Jul. 2026 - Present

- **Designed RISolate**, a framework for automatically deriving fine-grained RISC-V context-switch isolation policies from architectural/system documentation and historical security bugs across **U→U, S→S, VS→VS, and HS→VS** transitions.
- **Implementing automated policy-to-test translation** to generate executable isolation tests covering register/CSR, floating-point, vector, translation, and virtualization state.
- **Identified one reliability bug in OpenSBI and multiple security bugs across OpenSBI, Linux/KVM, Penglai, and ACE-RISC-V**; reported all findings to the respective maintainers.

Eliminating Tracing Overhead in LSM-Tree Based KV Stores: Workload Generation via ML

Role: First Author | Institut Teknologi Sepuluh Nopember (ITS) | Undergraduate Research | 2022 - 2023

- **Designed an offline ML-based workflow** for generating synthetic multi-level RocksDB traces to reduce reliance on runtime workload tracing.
- **Applied and compared multiple ML approaches**, including **linear/logistic regression, CTGAN, MLP, LSTM, and GRU**, using traces collected and processed from diverse RocksDB workloads.
- **Achieved a best similarity score of 75.19%** against organic traces using a custom regression- and classification-based evaluation metric.

EXPERIENCE

Arizona State University

Graduate Teaching Assistant, CSE 230: Computer Organization and Assembly Language | Tempe, AZ | Fall 2025

- Held office hours and helped undergraduate students understand **MIPS assembly language** and computer organization concepts.
- Debugged student MIPS programs involving registers, memory addressing, control flow, stack usage, and procedure calls, while grading assignments and supporting course exercises/labs.

The Software Practice Pte Ltd

Application Support Engineer | Singapore (Remote) | Apr. 2024 - Jul. 2024

- Maintained and supported mission-critical web and mobile applications, performing functional testing and investigating software issues to support ongoing client requirements.
- Prepared and validated project data in spreadsheets and databases, including data entry, annotation, and questionnaire-related tasks.

Institut Teknologi Sepuluh Nopember (ITS)

Laboratory Assistant, Database Management Systems | Surabaya, Indonesia | Jul. 2022 - Jun. 2023

- Assisted second-year students during database laboratory sessions and supported exercises using **Oracle SQL**.
- Helped students understand database creation, table structures, and core relational database concepts.

PUBLICATIONS

Chuqi Zhang, Spencer Faith, Feras Al-Qassas, **Theodorus Februento**, Zhenkai Liang, and Adil Ahmad.

“Fine-Grained Kernel Auditing using Augmented Syscall Reference Behavior Analysis and Virtualized Selective Tracing.” *IEEE Symposium on Security and Privacy (S&P)*, 2026.

- **Contributed to the design of APPARE’s LLM-assisted syscall reference behavior analysis**, combining dynamic execution profiles, static kernel call graphs, and code semantics to identify plausible unobserved control-flow transitions.
- **Developed the RAG-based code-semantics reasoning pipeline** that traverses kernel call graphs, retrieves historical execution context and function semantics, and uses an LLM to classify candidate control-flow edges as likely or unlikely.
- **Evaluated the LLM-assisted analysis**, showing that richer kernel semantic context improved code-inference accuracy by **over 14%**, helping APPARE construct more precise syscall reference behaviors for selective kernel auditing.

TECHNICAL SKILLS

Languages: Python, C, C++, RISC-V Assembly, MIPS Assembly

Systems: Linux, xv6, OpenSBI, KVM, QEMU, Spike, RVVM, Renode

Architecture & Security: RISC-V Privileged Architecture, Virtualization, Virtual Memory, PMP, TLB, Context Switching

Testing & Tools: Automated Testing, Conformance Testing, Symbolic Execution, KLEE, GDB, Git

LLM / AI: LLM-assisted Test Generation, Agentic Workflows, Retrieval-Augmented Generation (RAG)

RELEVANT COURSEWORK

Advanced Operating Systems (RISC-V, xv6); Software Verification, Validation & Testing; Current Topics in Cybersecurity; Foundations of Algorithms; Topics in Natural Language Processing (Transformers, LLMs)